

# CHUN-CHE WEN, PhD

Bayesian Biostatistician | Spatial and Network Methods | Health Services Research

Lebanon, NH | [chun-che.wen@dartmouth.edu](mailto:chun-che.wen@dartmouth.edu) | (678) 430-9283

[Website](#) | [Google Scholar](#) | [GitHub](#)

## ACADEMIC APPOINTMENT

**Postdoctoral Research Associate** | The Dartmouth Institute for Health Policy & Clinical Practice, Lebanon, NH

Oct 2024 - Present

Geisel School of Medicine at Dartmouth | Mentor: James O'Malley, PhD

## EDUCATION

**PhD in Biostatistics** | Medical University of South Carolina, Charleston, SC

Aug 2019 - Aug 2024

Dissertation: *Bayesian Beta-Binomial Mixed Models for Longitudinal, Clustered, and Spatiotemporal Data* | Advisor: Brian Neelon, PhD

**MS in Statistics** | Florida State University, Tallahassee, FL

Aug 2017 - May 2019

**BS in Applied Mathematics** | National Chung Hsing University, Taichung, Taiwan

Sep 2012 - Jun 2016

## RESEARCH INTERESTS

Bayesian hierarchical modeling; spatial and spatiotemporal statistics; statistical network analysis and community detection; longitudinal and mixture models; health-services and outcomes research.

## PEER-REVIEWED PUBLICATIONS

- Wen, C.-C., Paul, R., Hunt, K., Li, H., Hill, E., Malek, A., and Neelon, B. (2026). "A Bayesian zero-inflated spatiotemporally varying coefficients model for overdispersed binomial data." *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 189(3).
- Hunt, K., Wen, C.-C., Chundru, K., Mateus, J., Florez, H., Korte, J., Neelon, B., Wilson, D., Pearce, J., Alkis, M., Simpson, S., Alsbrook, E., Spate, V., and Malek, A. (2025). "Adverse pregnancy and infant outcomes by COVID-19 infection status before and during pregnancy." *Pregnancy*, 1, e70051.
- Malek, A., Wen, C.-C., Neelon, B., Wilson, D., Mateus, J., Pearce, J., Simpson, S., Chundru, K., Korte, J., Florez, H., and Hunt, K. (2024). "Prevalence of hypertensive disorders of pregnancy and chronic hypertension increased throughout the COVID-19 pandemic in South Carolina (2015-2021)." *Women & Health*, 1-15.
- Hunt, K., Wen, C.-C., Neelon, B., Wilson, D., Mateus, J., Pearce, J., Simpson, S., Chundru, K., Korte, J., Florez, H., and Malek, A. (2024). "Increasing prevalence of diagnosed gestational diabetes in South Carolina: 2015-2021." *Journal of Women's Health*, 33(11), 1518-1527.
- Simpson, S., Malek, A., Wen, C.-C., Neelon, B., Wilson, D., Mateus, J., Pearce, J., Chundru, K., Korte, J., Florez, H., and Hunt, K. (2024). "Trends in gestational weight gain and prepregnancy obesity in South Carolina, 2015 through 2021." *Preventing Chronic Disease*, 21, 240137.
- Wen, C.-C., Baker, N., Paul, R., Hill, E., Hunt, K., Li, H., Gray, K., and Neelon, B. (2024). "A Bayesian beta-binomial piecewise growth mixture model for timeline followback data." *Statistical Methods in Medical Research*, 33(10), 1859-1876.
- Hung, H.-Y., Wen, C.-C., Su, P.-F., Man, S., and Chang, Y.-J. (2024). "Implementation of shared decision-making about rooming-in: A before-and-after audit of breastfeeding in Taiwan." *International Breastfeeding Journal*, 19(1), 40.
- Chundru, K., Korte, J., Wen, C.-C., Neelon, B., Wilson, D., Mateus, J., Pearce, J., Simpson, S., and Hunt, K. (2024). "Increasing preterm delivery and small-for-gestational-age trends in South Carolina during the COVID-19 pandemic." *International Journal of Environmental Research and Public Health*, 21(4), 465.
- Wen, C.-C., Baker, N., Paul, R., Hill, E., Hunt, K., Li, H., Gray, K., and Neelon, B. (2023). "A Bayesian zero-inflated beta-binomial model for longitudinal data with group-specific changepoints." *Statistics in Medicine*, 43(1), 125-140.
- Neelon, B., Wen, C.-C., and Benjamin-Neelon, S. E. (2022). "A multivariate spatiotemporal model for tracking COVID-19 incidence and death rates in socially vulnerable populations." *Journal of Applied Statistics*, 50(8), 1812-1835.
- Killeen, T., Wen, C.-C., Neelon, B., and Baker, N. (2023). "Predictors of treatment completion among women receiving integrated treatment for comorbid posttraumatic stress and substance use disorders." *Substance Use & Misuse*, 58(4), 500-511.
- Wang, W., Wen, C.-C., Hung, T., Zhong, J., and Wen, M. (2022). "Single-stage sampling procedure for heteroscedasticity analysis of means." *Brazilian Journal of Probability and Statistics*, 36(4), 704-724.
- McGhee, H., Gehle, D., Shope, C., Wen, C.-C., Marston, A., Discolo, C., and Pecha, P. (2022). "Feeding performance and outcomes in infants with Robin sequence undergoing mandibular distraction osteogenesis." *The Cleft Palate-Craniofacial Journal*, 61(2), 295-301.
- Wang, W., Wen, C.-C., Das, R., and Wen, M. (2022). "A modified single-stage sampling procedure for heteroscedasticity analysis of means with unbalanced design." *Communications in Statistics - Simulation and Computation*, 53(5), 2277-2290.
- Wen, M., Wen, C.-C., and Wang, W. (2022). "Single-stage sampling procedure for heteroscedasticity in multiple comparisons with a control." *Communications in Statistics - Simulation and Computation*, 53(4), 1719-1728.
- Yan, F., Levy, D. A., Wen, C.-C., Melvin, C., Ford, M. E., Nietert, P. J., and Pecha, P. P. (2020). "Rural barriers to surgical care for children with sleep-disordered breathing." *Otolaryngology-Head and Neck Surgery*, 166(6), 1127-1133.
- Zhou, A., Wang, D., McKee, S., Ye, F., Wen, C.-C., Wallace, D. E., Ancell, K. K., Conry, R. M., and Johnson, D. B. (2019). "Correlates of response and outcomes with talimogene laherparepvec." *Journal of Surgical Oncology*, 120(3), 558-564.

## CONFERENCE PROCEEDINGS

- Sun, N., Young, T. C., Wen, M., Wen, C.-C., and Chin, C. (2024). "Stratified hypothesis testing of mean duration of response in randomized clinical trials." *Proceedings of the 2024 Joint Statistical Meetings*, Portland, Oregon. Zenodo.

## MANUSCRIPTS UNDER REVIEW

- A Bayesian spatial hierarchical model with multivariate random effects for quantifying multi-group regional heterogeneity. Under review at Biostatistics.
- Estimating the impact of hedge fund flows on liquidation risk. Under review.

3. Toward personalized medicine for smoking cessation: Predicting individuals' response to smoking cessation pharmacotherapy using multivariable and machine-learning models. Under review.
4. Melatonin supports early synaptic preservation or remodeling after cerebral ischemia: In vitro evidence for MT1/MT2-associated BDNF regulation. Revision in progress.

WORKING PAPER

1. "Bayesian community detection in asymmetric patient-flow networks."

SELECTED RESEARCH EXPERIENCE

- Research Assistant | Medical University of South Carolina, Charleston, SC

Aug 2021 - Aug 2024

Investigators: Kelly Hunt, PhD, and Angela Malek, PhD

Studied the effects of the COVID-19 pandemic, SARS-CoV-2 infection, and social determinants of health on pregnancy complications and maternal and infant outcomes.
- Research Assistant | Medical University of South Carolina, Charleston, SC

Aug 2019 - Aug 2024

Investigator: Brian Neelon, PhD

Developed multivariate spatiotemporal models for correlated count outcomes and applied them to daily county-level COVID-19 incidence and mortality.
- Summer Research Intern | Vanderbilt University Medical Center, Nashville, TN

May 2018 - Jul 2018

Supervisor: Fei Ye, PhD | Role: Data Analyst

TECHNICAL SKILLS

|                        |                               |
|------------------------|-------------------------------|
| Statistical computing  | R, SAS, Python                |
| Bayesian computing     | NIMBLE, Stan                  |
| Reproducible reporting | R Markdown, Bookdown, R Shiny |

PRESENTATIONS

INVITED PRESENTATIONS

- 2025

"A Bayesian zero-inflated spatiotemporally varying coefficients model for overdispersed binomial data." *Joint Statistical Meetings.*
- 2024

"Exploring the beta-binomial model: Analyzing overdispersed binomial data." *Hollings Cancer Center, Medical University of South Carolina.*
- 2024

"A Bayesian beta-binomial piecewise growth mixture model for timeline followback data." *Joint Statistical Meetings.*
- 2023

"A Bayesian zero-inflated beta-binomial model for longitudinal data with group-specific changepoints." *Eastern Chapter of the International Society for Bayesian Analysis Conference.*

CONTRIBUTED PRESENTATIONS

- 2025

"A Bayesian zero-inflated spatiotemporally varying coefficients model for overdispersed binomial data." *ENAR Spring Meeting.*
- 2024

"A Bayesian beta-binomial piecewise growth mixture model for timeline followback data." *WNAR/IMS/Graybill Annual Meeting.*
- 2024

"A Bayesian beta-binomial piecewise growth mixture model for timeline followback data." *SC-ASA Palmetto Symposium. Outstanding Presentation Award.*
- 2024

"A Bayesian beta-binomial piecewise growth mixture model for timeline followback data." *ENAR Spring Meeting.*
- 2023

"A Bayesian zero-inflated beta-binomial model for longitudinal data with group-specific changepoints." *Association of Clinical and Translational Statisticians Annual Meeting.*
- 2023

"A Bayesian zero-inflated beta-binomial model for longitudinal data with group-specific changepoints." *ENAR Spring Meeting.*
- 2023

"A Bayesian zero-inflated beta-binomial model for longitudinal data with group-specific changepoints." *Third Annual Data Science Workshop. Best 5-Minute Presentation Award.*

POSTER PRESENTATIONS

- 2023

"A Bayesian zero-inflated beta-binomial model for longitudinal data with group-specific changepoints." *Southern Regional Council on Statistics.*

TEACHING AND MENTORING

- Independent Study Journal Club | Dartmouth College, Lebanon, NH

Spring 2025

Bayesian Methods and Algorithms in Public Health
- Guest Lecturer - BMTRY 719 | Medical University of South Carolina, Charleston, SC

2021, 2024

Bayesian Biostatistics

Introduced Bayesian computation with NIMBLE and demonstrated MCMC implementation through applied case studies for biostatistics PhD students.
- Laboratory Instructor/Grader - BMTRY 701 | Medical University of South Carolina, Charleston, SC

Spring 2023

Biostatistical Methods II: Regression Methods in Biology and Medicine

Led weekly SAS laboratories on linear and logistic regression for MPH students.
- Laboratory Instructor/Grader - BMTRY 700 | Medical University of South Carolina, Charleston, SC

Fall 2021

Biostatistical Methods I: Introduction to Clinical Biostatistics

Taught SPSS and graded assignments for MPH students.
- Undergraduate Research Mentor | Statistics Undergraduate Research Experience Program, SRCOS

2023

HONORS AND AWARDS

2. Postdoctoral Scholar Professional Development Award, April 2025

- 3. Outstanding Presentation Award, SC-ASA Palmetto Symposium, April 2024
- 4. NSF Travel Award, Third Annual Data Science Workshop, March 2023
- 5. Best 5-Minute Presentation Award, Third Annual Data Science Workshop, March 2023
- 6. Travel Award, Southern Regional Council on Statistics, June 2023

**PROFESSIONAL SERVICE**

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- 2025**      Session Chair, Joint Statistical Meetings
- 2024**      Session Chair, ENAR Spring Meeting
- 2023**      Session Chair, Eastern Chapter of the International Society for Bayesian Analysis Conference

**JOURNAL REFEREEING**

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Biometrical Journal; Biostatistics; BMC Medical Research Methodology; Journal of Biopharmaceutical Statistics; Journal of Information Systems Education; Journal of Nonparametric Statistics; Journal of Statistical Theory and Practice; npj Science of Learning; PLOS ONE; Statistics in Medicine; Statistical Methods in Medical Research (5 reviews).